6,906: Economics: Macro for Computer Science

Lecture 1: Introduction

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Contact

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Course objectives

- This course aims to provide you with the key concepts and tools to understand recent developments in macroeconomics and assess their impact on the business environment and people's well-being.
  - Study how individuals and firms are affected by national and global events; recent examples include the global financial crisis, the COVID-19 pandemic, and the Russia - Ukraine war.
  - Analyze how the government, through monetary and fiscal policies, limits the costs imposed by these events on businesses and households.

Learning objectives

Factual knowledge:

1. Characterize a nation's economic health.
2. Evaluate the scope for policy to improve macroeconomic performance.
3. Use simple macroeconomic models and run numerical simulations of these models to study economic phenomena.
4. Demonstrate a deep analytical understanding of the role of expectations for economic decisions.

Learning objectives

Practical ability:

1. Access, critically analyze, evaluate, and synthesize quantitative and qualitative economic data.

Transfer knowledge:

1. Present research findings orally and in writing clearly and concisely to diverse audiences.
2. Contribute constructively to team project work.

Course content

- Whereas microeconomics studies the decision-making of small economic units (e.g., individuals, households, and firms), macroeconomics focuses on the functioning of the whole economy.
- A good understanding of how the overall economy works can help you spot cyclical trends, such as expansions and recessions, and anticipate economic developments and political decisions. This allows for taking informed action and making financial decisions accordingly.

Course content

The study of modern macroeconomics is usually divided into three "runs": the long run, the medium run, and the short run, each being discussed in a part of this course.
The course comprises four parts focused on:

1. Introduction of key macroeconomic terms such as output, inflation, and employment.
2. An economy's performance over decades by laying out the main facts about long-run growth and then developing a model that describes how capital accumulation and technological progress determine economic growth.
3. The economy's behavior in the short run, spanning from a couple of months to a few years, and derives the workhorse model used to study short-run movements in output.
4. The medium run, measured in periods of several years, in which the relationship between inflation and unemployment is discussed, and the short-run model is enlarged to incorporate this relationship as well.

Course material and literature

- The economic theory we will cover is partly developed in the following books:
  - **Mandatory:**
    Olivier Blanchard, 2021. Macroeconomics, Global Edition, 8th edition, Pearson, available at https://hsg.swisscovery.slsp.ch/permalink/41SLSP_HSG/1h7ra77/alma9911065107605506
  - **Supplementary:**
    1. Julio Garin, Robert Lester, and Eric Sims, 2021. Intermediate Macroeconomics, v. 3.0.1, unpublished, available at https://juliogarin.com/files/textbook/GLS_Intermediate_Macro.pdf
    2. Peter Birch Sorensen and Hans Jorgen Whitta-Jacobsen, 2022. Introducing Advanced Macroeconomics, 3rd edition, Oxford University Press.
- Other references will be provided in class.

Course material and literature

- All course material (lecture slides, codes, exercises, quizzes, solutions, etc) is posted on the Canvas/Studynet page of the course.
- The mandatory textbook (i.e. Blanchard, 8th edition) adds to the lecture material but is no substitute.
- The listed chapters in the textbook are compulsory reading, but only the parts discussed in class.
- Only the material discussed in the lectures is part of the examination.

Course organization

- The course is organized into 12 weekly sessions of 90 minutes.
- All sessions are held on campus.
- The course is not divided into lectures and exercise sessions (tutorials); each session is a mix of theory and applications.

Course structure

Lecture no	Lecture title	Date	Book chapters
1	Introduction to Macroeconomics	20.02	Blanchard: Chapter 2 Sorensen & Whitta-Jacobsen: Chapter 1
2	Key Macroeconomic Concepts	27.02	Blanchard: Chapter 2, 3, 7 Garin et at.: Chapter 1 Sorensen & Whitta-Jacobsen: Chapter 10,11,12
3	Facts about growth	6.03	Blanchard: Chapter 10 Garin et al.: Chapter 4, 7 Sorensen & Whitta-Jacobsen: Chapter 2
4	Economic growth and technological progress	13.03	Blanchard: Chapter 11, 12 Garin et al.: Chapter 5, 6 Sorensen & Whitta-Jacobsen: Chapter 3, 5
5	Facts about business cycles	20.03	Garin et al.: Chapter 19 Sorensen & Whitta-Jacobsen: Chapter 13
6	Goods and financial markets	27.03	Blanchard: Chapter 3, 4
7	Short-run economic fluctuations and the IS-LM model	17.04	Blanchard: Chapter 5,6 Garin et al.: Chapter 24
8	Inflation, unemployment, and the Phillips curve	24.04	Blanchard: Chapter 7,8,9
9	Expectations and policy	1.05	Blanchard: Chapter 16
10	Revision, Q&A	8.05	
11	Exam	15.05	
12	Presentations of projects	22.05	

Course grade

- Intermediate examination (mandatory, group): 40%
  - Group project (maximum three students) with a 15-minute presentation in class on 22.05.
  - Students will be assigned a group of countries (one for each student) for which they have to collect data, derive stylized facts, implement and simulate the models studied in class, and report the results in an oral presentation, accompanied by computer codes and any other supporting materials.
  - Students may choose any program they want (e.g., MATLAB, Python, R).
- Final examination (mandatory, individual): 60%
  - Written final exam during lecture time on 15.05.
  - Duration of 90 minutes, based on all the material covered in the course.
- Allowed: One A4 double-sided cheat sheet and a non-programmable calculator

The microeconomic approach

- **Microeconomics** is the branch of economics that studies the decision-making of individuals (e.g., people, firms) regarding the allocation of scarce resources and the prices at which they trade goods and services.
- **Important concepts**:
  - Scarcity: limited resources and infinite wants
  - Opportunity cost: the value of the next-best alternative
  - Cost-benefit analysis: the general rule is to consume where marginal benefit = marginal cost
  - Marginal benefit: the benefit from consuming an additional unit of a good or service
  - Marginal cost: the cost of consuming an additional unit of a good or service

The microeconomic approach

Demand and supply

- **Demand** is the amount of product or service that consumers are willing to purchase at a particular price level, and is influenced by:
  - Type of good/service (essential, inferior, etc.)
  - Consumer's ability to pay (income) and tastes
  - Availability/price of complementary goods and substitutes
  - Number of buyers
- **Supply** is the amount of product or service that producers are willing to provide at a particular price level, and is influenced by:
  - Availability of resources used as inputs in production
  - Capital/labor intensity and inputs' prices
  - Technology
  - Entry/exit barriers and numbers of firms
- **Elasticity** measures a variable's sensitivity to a change in another variable (e.g., price elasticity of demand is the change in consumer demand due to a change in a good or service's price)

The microeconomic approach

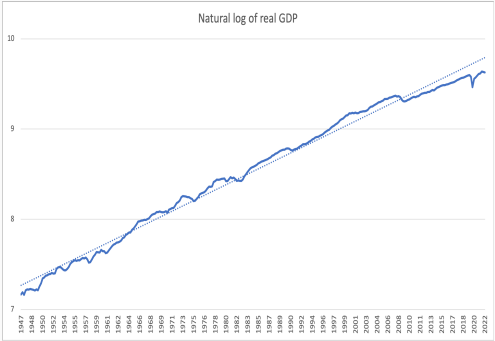
Pricing of a good or service

- Some factors that influence price determination are:
  - Marginal benefit of consumers and marginal cost of production
  - Government and legal regulations
  - Relative scarcity of a good or service with respect to its demand
  - Extend of competition in the market, market structure, and market power
  - Pricing objectives (e.g., profit maximization, obtaining market share leadership)
  - Marketing methods (e.g., quality of salesforce)

The macroeconomic approach

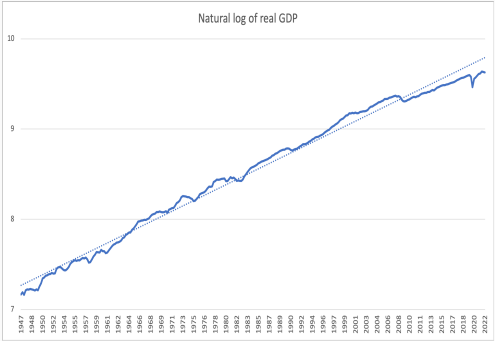
- **Macroeconomics** is the branch of economics that studies the structure, performance, and decision-making of the whole aggregate economy.
- Among the variety of macroeconomic concepts, **output**, **inflation**, and **unemployment** are key.

Output



- Output or gross domestic product (GDP) measures the total market value of all domestic goods and services produced in a given period (e.g., year).

Output



- When compared with prior periods, GDP tells us whether the economy is expanding by producing more goods and services or contracting due to less output.
- The economy's movement through these alternating periods of expansion and contraction is known as the **business cycle**.
- The **economic growth rate** of an economy is expressed as a percentage that shows the rate of change in its GDP, typically from one year to the next.

Inflation

- **Inflation** is a concept that refers to increases in the price level of goods over a given period.
- Stable prices are crucially important for an economy.
- Substantial price fluctuations over longer periods lead to considerable uncertainty between companies and households, as well as a redistribution of wealth and income. This affects everyone's level of prosperity.
- For example, the Swiss National Bank considers prices in Switzerland to be stable when the overall price level increases by less than 2% per year, and there is no incidence of negative inflation (deflation).

Unemployment rate

- The **unemployment rate** is calculated as the percentage of the labor force that is unemployed.
- Low inflation and full employment are the cornerstones of monetary policy for the modern central bank.
- However, there is an inverse relationship between unemployment and inflation (Phillips curve).
  - When unemployment is low, workers are hard to find, so employers raise wages, leading to higher prices (inflation).
- At every moment, central bankers face a trade-off. They can stimulate production and employment at the cost of higher inflation. Or they can fight inflation at the cost of slower economic growth.

Summary

- In this lecture, we learned:
  - To differentiate between a micro- and macro-approach to economic analysis.
- In the next lecture, we will study several key macroeconomic concepts.